LAB 09

# Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

| Item | Detail |
|---|---|
| Topic | Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca |
| Pipeline stage | Evaluation |
| Duration | 25 min |
| Simulator page | http://localhost:8000/#/analytics |
| Mode | No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only. |

*Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.*

## Scenario

The investment committee wants to know which of the pipeline's signals deserves capital. Return alone cannot answer that, so the evaluation stage runs every signal through the same backtester, over the same window, with the same costs, and compares it with simply holding SPY.

## Objective

Read the full metrics table and compare a strategy with buy-and-hold.

## Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 09. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

## Lab Steps - In the Trading Simulator

*No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 09: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.*

### Step 1. Open the analytics

Analytics: choose the EMA 10/30 seeded backtest. Hover every metric for its formula.

Open: http://localhost:8000/#/analytics (Analytics)

*Check: You can define Sharpe, Sortino, Calmar, max drawdown, profit factor, expectancy, exposure and turnover.*

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### Step 2. Compare with buy-and-hold

Tick the EMA backtest and 'Buy and hold SPY' and press Compare.

Open: http://localhost:8000/#/analytics

*Check: Buy-and-hold has higher CAGR but a deeper drawdown; the strategy has lower exposure.*

### Step 3. Glossary

Open Help & Glossary > Metric glossary.

Open: http://localhost:8000/#/help?tab=glossary

*Check: Each formula matches the course slides.*

**Expected in the app: A metrics table you can explain line by line.**

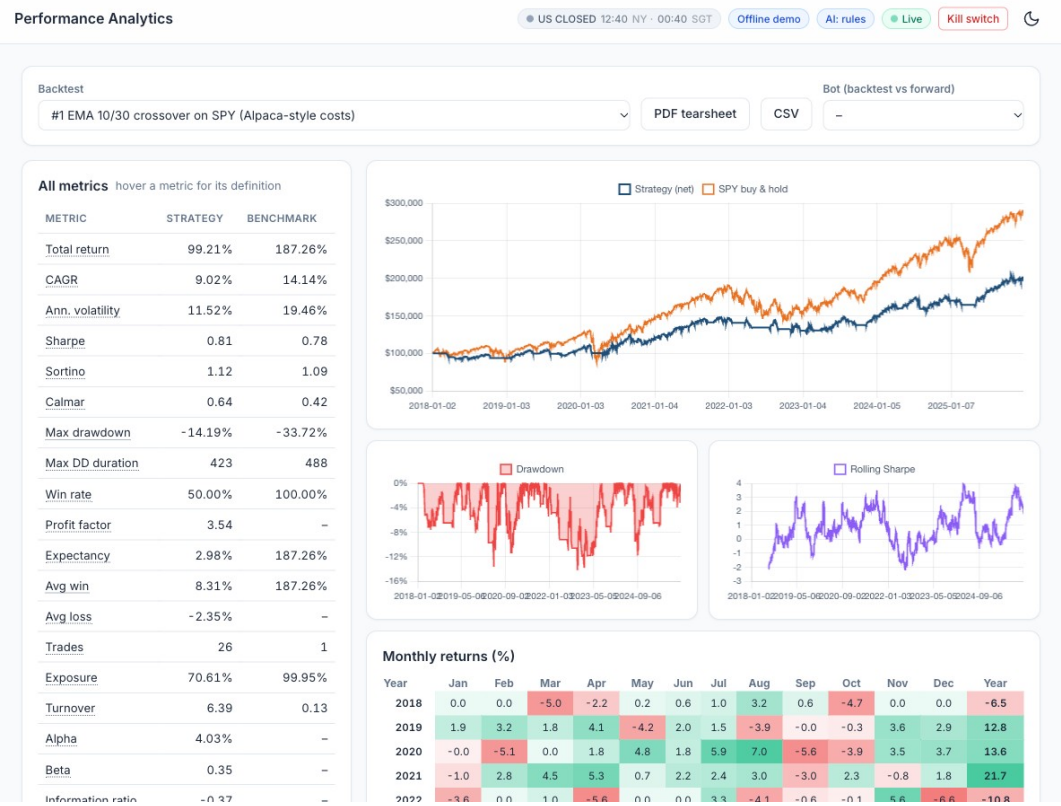



*Figure: Lab 09 in the Trading Simulator.*

## Optional: Let AI Write the Code

*This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.*

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| Performance and risk metrics against buy-and-hold - run it in Google Colab |
| --- |
| I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: for an EMA 10/30 crossover on SPY 2018-2025 (trade at the next day's open with a 0.05% cost per trade) compute and explain total return, CAGR, annual volatility, Sharpe, Sortino and Calmar ratios, maximum drawdown, win rate, profit factor and exposure, side by side with buy-and-hold SPY, and plot both equity curves. |

Open in Claude | Open in ChatGPT | Open Google Colab

## Debrief

Performance metrics answer different questions: CAGR measures growth, Sharpe and Sortino reward per unit of risk, drawdown and Calmar the pain an investor must survive, beta how much is just the market. Running every strategy through one evaluation stage is what makes the comparison fair.

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